

USE CASE: LOG GRANT EXPENSE

SRN: PES1UG24CS437

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Actor: Faculty Researcher

Preconditions:

- Faculty Researcher is authenticated and logged into the system.
- Faculty Researcher has at least one active, approved grant assigned to them.

Postconditions:

- The expense is recorded against the grant's ledger.
- The grant's remaining budget balance is updated to reflect the new expense.
- An immutable audit log entry is created for the transaction (as per NFR-001).

Main Success Scenario:

1. Faculty Researcher selects an active grant from their dashboard.
2. Faculty Researcher selects "Log Expense" and enters expense details.
3. The system validates the input fields for completeness and correct format.
4. The system checks the expense amount against the grant's remaining budget balance (include: Check Budget Balance).
5. The system confirms the expense does not exceed the remaining balance.
6. The system records the expense, updates the running budget balance, and writes an audit log entry.

7. The system displays a confirmation with the updated balance to the Faculty Researcher.

Alternate Flow:

- 4a. At step 5, the system detects the expense would cause the balance to go negative.
- 4b. The system blocks the transaction and displays an error indicating the expense exceeds the remaining budget.
- 4c. The system notifies the Research Dean that an override may be required.
- 4d. If the Research Dean approves an override, the system re-attempts step 6 with the override flag set and logs the override action separately in the audit trail.
- 4e. If no override is granted, the expense remains unrecorded and the Faculty Researcher is returned to step 2.

Exception Flows:

Invalid input data:

- 3a. At step 3, the system detects malformed or missing required fields.
- 3b. The system rejects the submission and highlights the invalid fields with a specific error message.
- 3c. The faculty researcher corrects and resubmits as per step 2.

System/Database Failure during Write:

- 6a. At step 6, the system encounters an error while writing the expense record or audit log entry.
- 6b. System aborts the transaction and rolls back any partial changes to the budget balance.
- 6c. System displays an error indicating the expense was not saved and logs the failure for administrator review.

6d. The faculty researcher may start over from step 2.

Concurrent Modification Conflict:

4a. At step 4, another transaction has modified the grant's balance since the faculty researcher loaded the page.

4b. The system detects the stale balance read and rejects the submission instead of risking an incorrect overdraft check.

4c. The system prompts the faculty researcher to refresh the grant view and resubmit with the current balance.

Session Expiry Mid-Submission:

2a. At step 2, the faculty researcher's authenticated session expires before submission is completed.

2b. The system detects the invalid/expired session on submission attempt and discards the unsaved entry.

2c. The system redirects the faculty researcher to the login page.

2d. The faculty researcher re-authenticates and must re-enter the expense from step 2

Supporting Document Upload Failure

2a. At step 2, the faculty researcher attaches a supporting document, but the upload fails.

2b. The system rejects the attachment and displays an error, while preserving the other entered field values.

2c. The faculty researcher retries the upload or removes the attachment and resubmits from step 2.